DHCP (Dynamic Host Configuration Protocol)

Purpose of DHCP

DHCP automatically assigns network configuration to devices when they join a network.
Without DHCP, every device would need manual configuration.

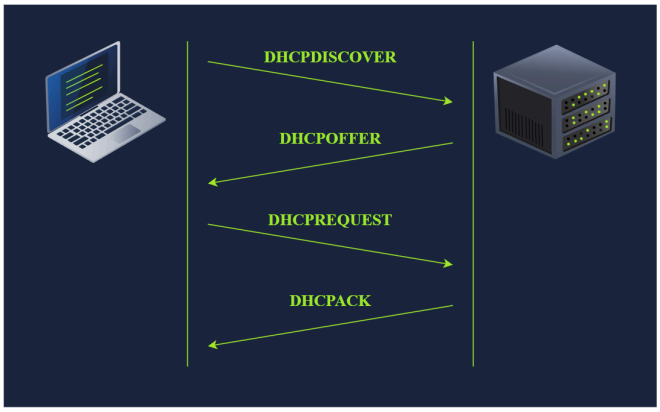
Network settings required to access a network

- IP Address
- Subnet Mask
- Default Gateway (Router)
- DNS Server

DHCP provides these automatically.

DHCP Process (DORA)

DORA = Discover → Offer → Request → Acknowledge



1. DHCP Discover

- Client searches for DHCP server.
- Broadcast message.

Source IP: 0.0.0.0
Destination IP: 255.255.255.255

2. DHCP Offer

- DHCP server responds.
- Offers an available IP address and configuration.
- Includes:
    - IP address
    - subnet mask
    - gateway
    - DNS server

3. DHCP Request

- Client accepts the offer.
- Broadcast message requesting the offered IP.

4. DHCP Acknowledge (ACK)

- Server confirms IP assignment.
- Client can now use the network.

ARP (Address Resolution Protocol)

Purpose of ARP

ARP is used to find the MAC address of a device when its IP address is known.

IP Address → MAC Address

When one host wants to send data to another host on the same network, it must:

- Know the destination IP address
- Find the destination MAC address

Even if a device knows the target IP address, it cannot send an Ethernet frame without the MAC address.

A MAC address is:

- 48-bit address
- Written in hexadecimal
- Example:
  - 7C:DF:A1:D3:8C:5C
  - 44:DF:65:D8:FE:6C
- Used in:
  - Ethernet (IEEE 802.3)
  - WiFi (IEEE 802.11)

Ethernet Frame Structure

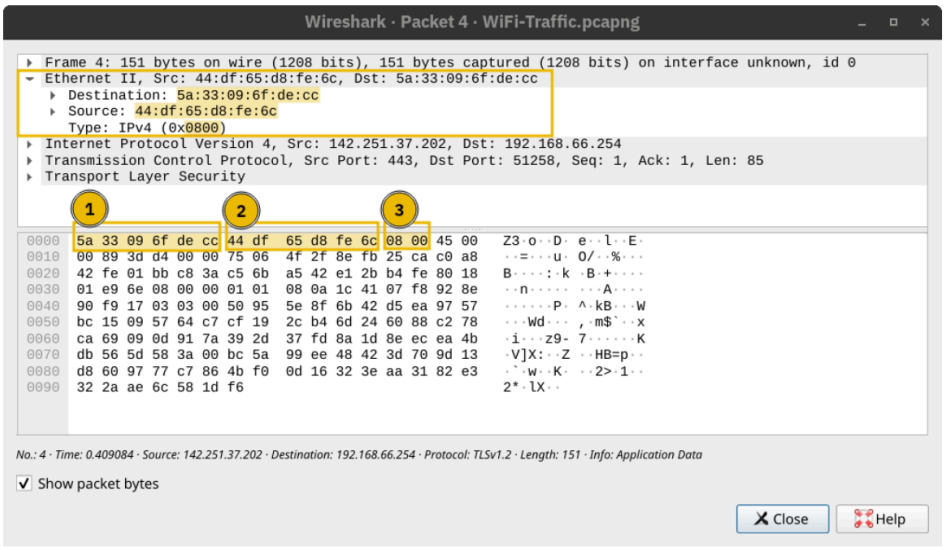
When sending data on a local network, the IP packet is encapsulated inside an Ethernet frame.

Ethernet header contains:

Field	Description
Destination MAC	Target device
Source MAC	Sender
Type	Protocol (IPv4, IPv6, ARP etc.)

Structure:

```
Ethernet Frame
├── Destination MAC
├── Source MAC
├── Type
└── IP Packet
```



ARP Communication Process

Example:
Host A wants to communicate with:
IP: 192.168.66.1
but only knows the IP, not the MAC.

Step 1 — ARP Request
Host sends a broadcast request:

```
Who has 192.168.66.1?
Tell 192.168.66.89
```

Broadcast means every device on the network receives it.

Step 2 — ARP Reply
The target host responds:

```
192.168.66.1 is at 44:df:65:d8:fe:6c
```

This message is unicast (sent directly to the requester).
Now the sender stores the MAC address in its ARP table.

ICMP (Internet Control Message Protocol)

Purpose of ICMP
ICMP is used for network diagnostics and error reporting.

It helps devices:
- check connectivity
- report network errors
- troubleshoot routing problems

ICMP operates at the Network Layer (Layer 3) and is carried inside IP packets.

Ping Command
Ping checks whether a target system is alive and reachable.

It also measures Round-Trip Time (RTT).
RTT = time for packet to go to target + return

How Ping Works

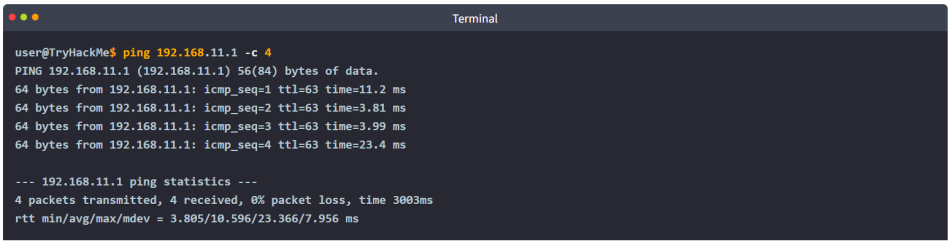
Ping uses two ICMP message types:
Echo Request → Type 8 → Sent to target
Echo Reply → Type 0 → Response from target

Process:

- Host A → ICMP Echo Request → Host B
- Host B → ICMP Echo Reply → Host A

Command:

```
ping 192.168.11.1 -c 4
```



Explanation:

-c 4 → send 4 packets only

No reply may happen if:

- target device is offline
- firewall blocks ICMP
- network routing issue
- ICMP disabled

Traceroute

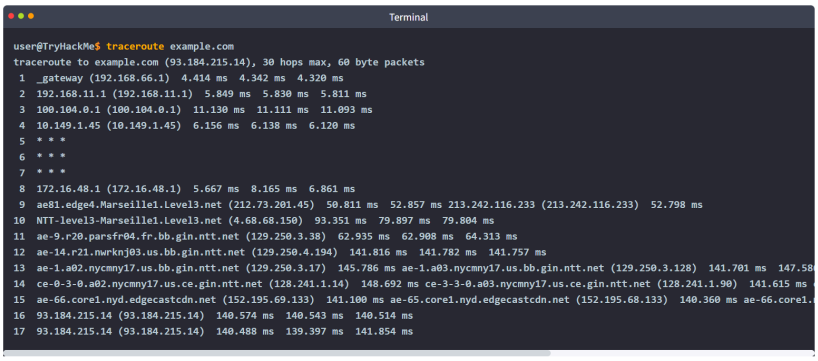
Traceroute discovers all routers between your system and the destination.

Linux / macOS:

```
traceroute
```

Windows:

```
tracert
```



How Traceroute Works

1. Traceroute uses the TTL (Time To Live) field in the IP header.
    a. TTL = maximum number of routers a packet can pass.
2. Each router:
    a. TTL = TTL - 1
3. If TTL becomes 0:
4. router drops the packet
5. router sends ICMP Time Exceeded message

Network Address Translation (NAT)

1. Why NAT Exists

IPv4 uses 32-bit addressing.
2^32 = 4,294,967,296 addresses

Due to the massive growth of Internet devices (computers, phones, IoT), IPv4 address exhaustion became a problem.

Solution:
Use NAT so multiple devices can share a single public IP address.

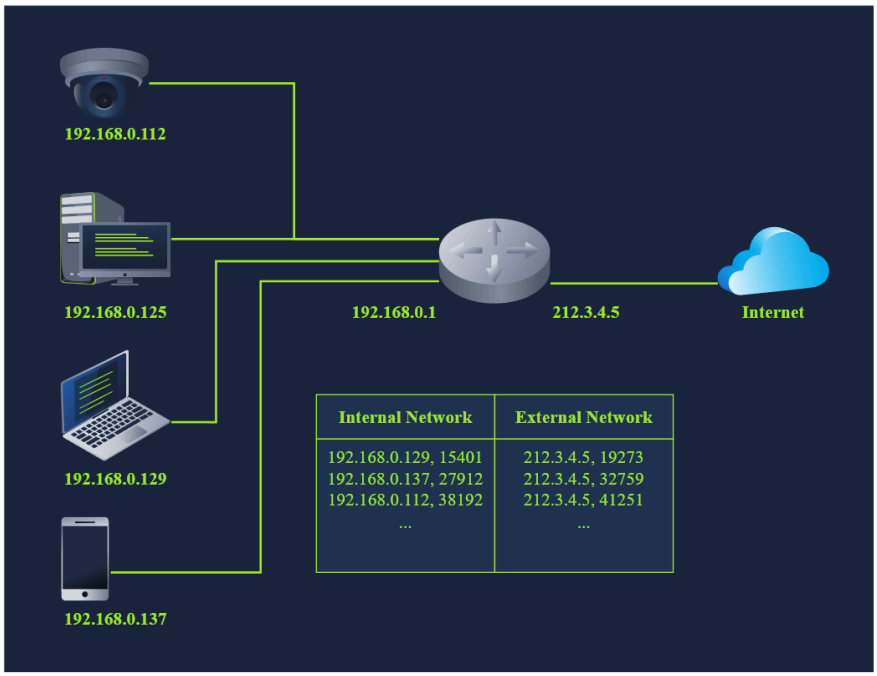
2. What is NAT?

Network Address Translation (NAT) is a technique used by routers to translate private IP addresses into public IP addresses when accessing the Internet.

Device	Private IP
Laptop	192.168.0.129
PC	192.168.0.130
Phone	192.168.0.131

Router Public IP: 212.3.4.5

All devices access the Internet through one public IP.



5. How NAT Works

1. Device sends request

Source: 192.168.0.129:15401
Destination: Web Server

2. Router performs translation

Router changes source address:
192.168.0.129:15401
↓
212.3.4.5:19273

3. Server response

Server replies to:
212.3.4.5:19273

4. Router checks NAT table

Router maps response back to:
192.168.0.129:15401

Packet is delivered to the laptop.